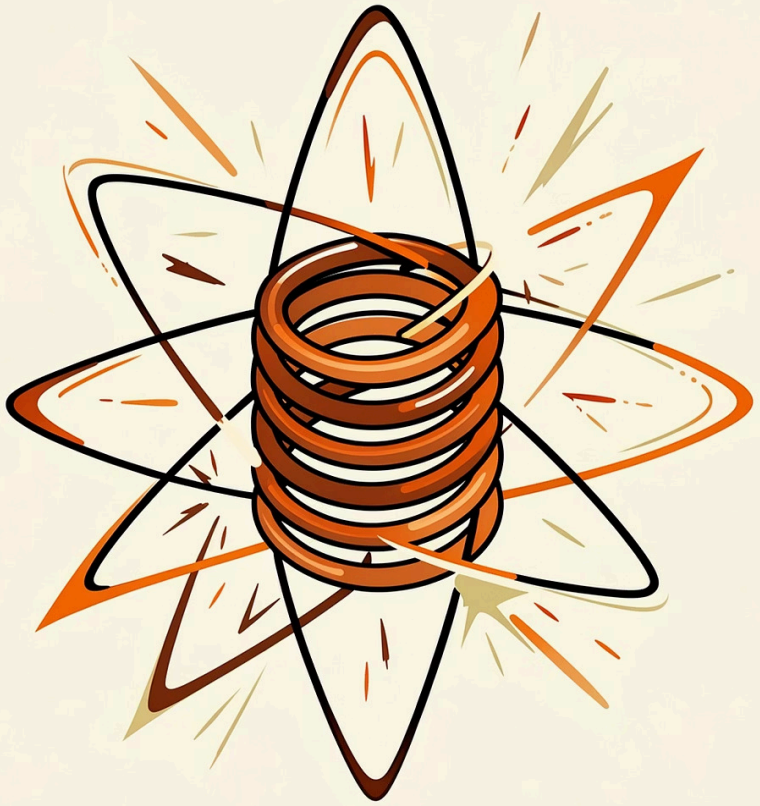
Magnetism & Electromagnetic Induction

From invisible forces to the engines of modern civilization — understanding how moving magnets and changing fields power our world.

B.TECH PHYSICS

ELECTROMAGNETISM



Made with GAMMA

The Dawn of Electromagnetism

For centuries, electricity and magnetism were treated as entirely separate forces of nature. That changed dramatically in the early 19th century.

Oersted's Discovery (1820)

A current-carrying wire deflects a compass needle — moving charges create magnetic fields.

Ampère's Law

Mathematical formulation of the relationship between electric currents and the magnetic fields they produce.

The Big Question

If currents create magnetism, can moving magnets create currents?

Made with GAMMA

~1830

Faraday & Henry: The Breakthroughs

Working independently on opposite sides of the Atlantic, **Michael Faraday** (England) and **Joseph Henry** (USA) made one of the most consequential discoveries in physics.

> "What is the use of a new-born baby?" — Faraday's reply when asked about the practical value of his discovery.

Electromagnetic Induction

A **changing magnetic field** induces an electric current in a closed coil — the foundation of all modern electrical power.

Made with GAMMA

Faraday's Experiments: Seeing Induction in Action

Faraday moved a bar magnet near a coil connected to a galvanometer and observed something extraordinary.

1 Current Only When Moving

The galvanometer deflects *only* when the magnet is in motion — a stationary magnet produces nothing.

2 Direction Reverses

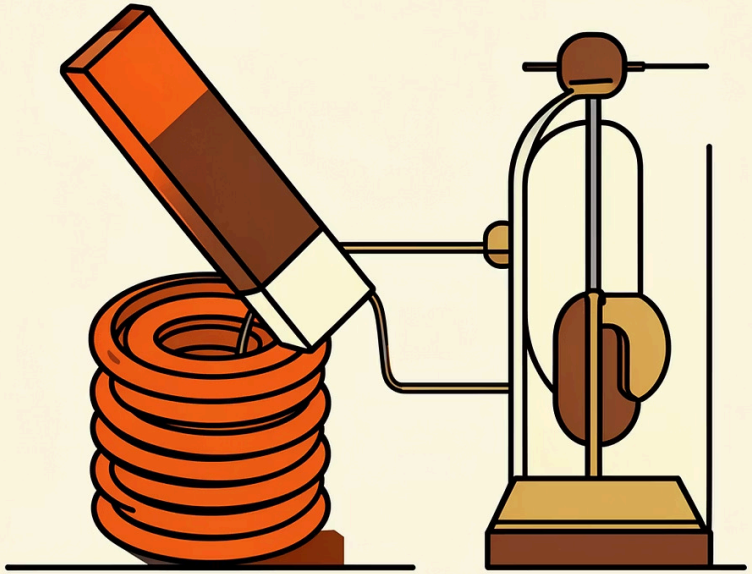
Pulling the magnet away reverses the current direction, confirming the role of *change* in flux.

3 Faster Motion = Larger Current

The rate of change of magnetic flux determines the induced EMF — the faster the motion, the greater the current.

Key Insight: It is the **relative motion** between magnet and coil — the change in magnetic flux — that drives induction.

Made with GAMMA



Faraday's Experiment 6.1

Motion Creates Current

A bar magnet approaching or receding from a coil induces a measurable EMF — the cornerstone demonstration of electromagnetic induction.

Made with GAMMA

The Power of Induction: From Lab to Life

Faraday and Henry's work laid the foundation for modern technology. Imagine a world without electricity — no lights, trains, or computers.

Generators

Convert mechanical energy into electrical energy through rotating magnets and coils.

Transformers

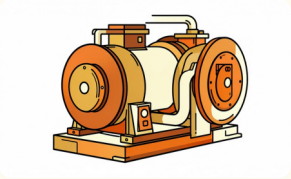
Step voltage up or down for efficient long-distance power transmission.

The Grid

Every power plant and electrical device relies on the principles of induction.

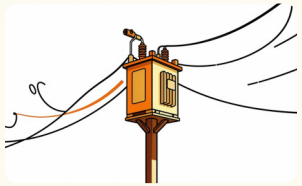
Made with GAMMA

Real-World Applications: Where Induction Powers Us



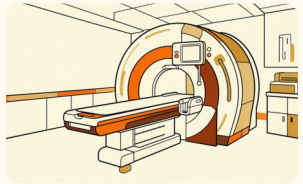
Electric Generators

Spinning magnets within coils induce current, powering homes, cities, and industries worldwide.



Transformers

Essential for stepping voltage up for transmission and down for safe consumer use.



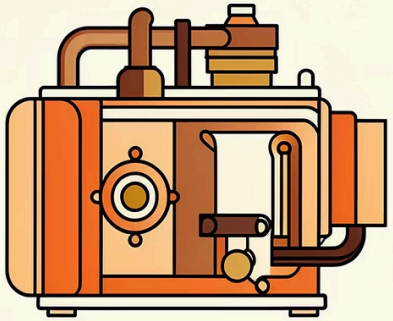
MRI Machines

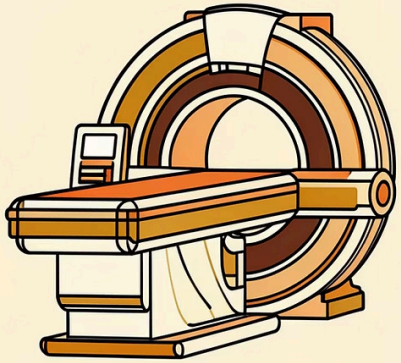
Powerful magnetic fields and induced radio signals create detailed anatomical images for diagnosis.

Microphones

Sound waves move a coil within a magnetic field, inducing electrical signals that become audio.

Made with GAMMA





Induction in Action

From Power to Precision

Whether generating gigawatts for a city or capturing millimeter-accurate medical images, electromagnetic induction is the invisible force at work.

Made with GAMMA

B.Tech Relevance: Core Principles & Formulas

1

Faraday's Law

Induced EMF is proportional to the **rate of change of magnetic flux**: $\mathcal{E} = -N\frac{d\Phi}{dt}$

2

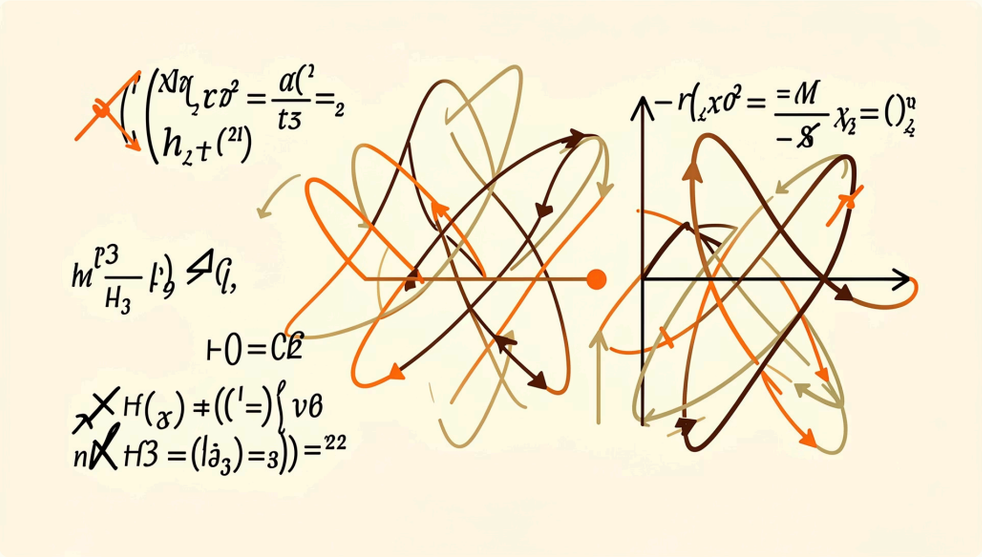
Lenz's Law

The induced current **opposes** the change in flux that produced it — nature resists change.

3

Self & Mutual Induction

Self-induction governs inductors; mutual induction underpins **transformers** and coupled circuits.



Made with GAMMA

The Future is Electric — Driven by Induction

Electromagnetic induction is not merely a textbook concept. It is the **engine of technological progress** — from renewable energy to advanced medical systems.

🌿 Renewable Energy

Wind turbines and hydroelectric generators rely entirely on induction principles.

🏥 Advanced Medicine

Next-generation imaging and therapeutic devices build on Faraday's legacy.

⚡ Innovation

Understanding induction is the key to designing the electrical systems of tomorrow.

Made with GAMMA